

The empirical recurrence for OEIS A232320, recovered from its tail

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Abstract

OEIS A232320 records “Empirical recurrence of order 78” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 230$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 83$. Its last coefficient is $c_{78} = 90112 \neq 0$, so no further tail satisfies anything shorter; any order-78 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232320 is “Number of $(4+1)X(n+1)$ 0..1 arrays with every element equal to some horizontal or antidiagonal neighbor, with top left element zero.”. It has offset 1 and begins

34, 551, 10409, 231454, 5142441, 113293694, 2499669158, 55147236337, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:18:40, revision 10) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 230$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 230$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 78: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 78.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 460$ exact terms from $n = 5$ on, it returns order 78 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	19	c_{21}	−3551469240197	c_{41}	−2481230313530186	c_{61}	427929588291
c_2	131	c_{22}	6648547263874	c_{42}	117119468067428	c_{62}	−139970935196
c_3	−1142	c_{23}	16749507209291	c_{43}	1364104236170232	c_{63}	−144277142921
c_4	−7590	c_{24}	−19871616002172	c_{44}	180303594529563	c_{64}	14424165730
c_5	32276	c_{25}	−64462946527517	c_{45}	−782250627773481	c_{65}	39548233295
c_6	248086	c_{26}	44675445035601	c_{46}	−242229057705391	c_{66}	1891454145
c_7	−529044	c_{27}	204029449169966	c_{47}	424709261219838	c_{67}	−8800558562
c_8	−5355129	c_{28}	−64587625994227	c_{48}	169657132122316	c_{68}	−1489465512
c_9	4971739	c_{29}	−533386659719721	c_{49}	−181362816047516	c_{69}	1578636236
c_{10}	82943604	c_{30}	8645320781385	c_{50}	−103018770926330	c_{70}	450274842
c_{11}	−12365251	c_{31}	1163694952177850	c_{51}	58175475453692	c_{71}	−215353780
c_{12}	−969506672	c_{32}	245791640778500	c_{52}	56403516490251	c_{72}	−93622308
c_{13}	−400117373	c_{33}	−2153086562232808	c_{53}	−13246587349907	c_{73}	19101488
c_{14}	8830849352	c_{34}	−684981715442138	c_{54}	−25553296023077	c_{74}	14887760
c_{15}	7578657317	c_{35}	3353722182895904	c_{55}	544839186590	c_{75}	−762944
c_{16}	−63955894173	c_{36}	914503954656811	c_{56}	9531648889432	c_{76}	−1654016
c_{17}	−79755609356	c_{37}	−4247568320439817	c_{57}	1517203406312	c_{77}	−4096
c_{18}	372756557124	c_{38}	−364838862006203	c_{58}	−2977330688198	c_{78}	90112
c_{19}	604236439576	c_{39}	3834544099182118	c_{59}	−1004298170040		
c_{20}	−1755729004819	c_{40}	−210673637724706	c_{60}	751204009271		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 83$, and it is the recurrence OEIS A232320 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{78} - c_1 x^{77} - \dots - c_{78}$. Its constant term is $-c_{78} = -90112$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 78 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 78, so $\deg q = 78$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 78, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 78, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 90112.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-78 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A232320.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.